

JKBOSE 2026

CLASS 11TH

#XUV ♥

EQUILIBRIUM

LECTURE-~~85~~ CHEMISTRY

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Bhaichara
ON
TOP



JK Bole Wallah

Topics To Covered

1

Ionic Equilibrium

2

Concept of Acids and Bases

REVISION.

Le-Chatelier PRINCIPLE

Factors affecting equilibrium.

a) EFFECT OF CONC.

Reactant conc $\uparrow$, $\rightleftharpoons$ Product side

Product conc $\uparrow$, $\rightleftharpoons$ backward direction.

b) Effect of Temp.

$T \uparrow$, $\rightleftharpoons$ Endothermic

$T \downarrow$, $\rightleftharpoons$ Exothermic.

c) Effect of Pressure.

$P \downarrow$ no. of moles of gas.

$P \uparrow$, $\rightleftharpoons$ no. of moles $\downarrow$ i.e.
(no. of moles $\uparrow$)

$P \downarrow$, $\rightleftharpoons$ no. of moles $\uparrow$
(no. of moles of gas $\downarrow$)

d) Effect of adding inert gas

↓
Const. Volume

No change in
equilibrium.

↓
Const. P.

Volume ↑
no. of moles of
gas per unit volume ↓
= brium → no. of moles of gas ↑.

INDUSTRIAL APPLICATIONS OF LE-CHATLIER.

PRINCIPLE.

- a) In production of ammonia (NH_3) by Haber's process.
- b) In production of sulphuric acid (H_2SO_4) by contact process.

Ionic Equilibrium



If the equilibrium exist between the ions in the solution, then it is known as ionic equilibrium.



But not all substances give ions in H_2O .

On the basis of dissociation of substances into ions, substances are divided into two substances. (Michael Faraday)

1) NON-ELECTROLYTES $\rightarrow$ Substances that do not produce ions in water are called Non-electrolytes. e.g. Urea, Glucose, Fructose, Sucrose etc.

2) ELECTROLYTES $\rightarrow$ Substances that produce ions in water are called Electrolytes e.g. All acids, bases and Salt (HCl , H_2SO_4 , NaOH , $\text{Mg}(\text{OH})_2$, NaCl , KCl).

On the basis of extent of dissociation into ions, electrolytes are further classified into two types.

a) Strong Electrolytes $\rightarrow$ Type of electrolyte

that show complete dissociation in water. are called strong electrolyte. e.g.

Acids $\rightarrow$ HCl , HI , H_2SO_4 , HNO_3 etc.

Base $\rightarrow$ NaOH , $\text{Mg}(\text{OH})_2$, KOH etc.

Salt $\rightarrow$ NaCl , KCl , MgCl_2 etc.

Ionic Equilibrium



b) WEAK ELECTROLYTE $\rightarrow$

Type of electrolytes that do not show complete dissociation in water i.e. incomplete dissociation.
e.g.

Acid $\rightarrow$ HF , CH_3COOH , H_2CO_3 , H_3BO_3 etc.

Base $\rightarrow$ NH_4OH , $\text{Ca}(\text{OH})_2$ etc.

Concept of Acids and Bases

OLD CONCEPT →

ACID → Substances that are sour in taste and convert blue litmus into red. were called acids.

BASE → Substances that are bitter in taste & convert red litmus into blue.



MODERN CONCEPT →

a) ARHENIUS CONCEPT.

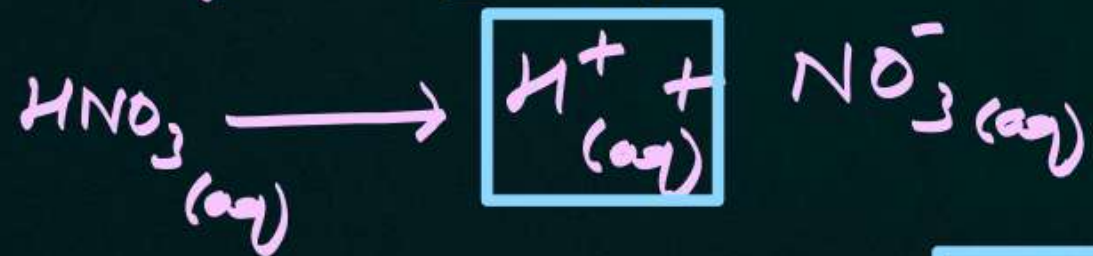
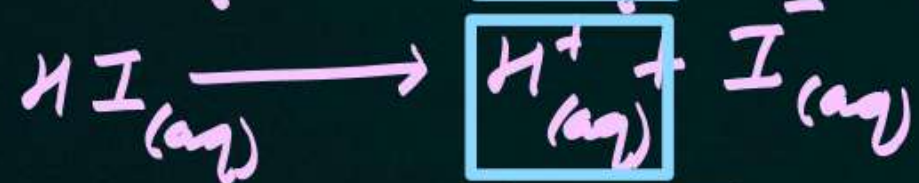
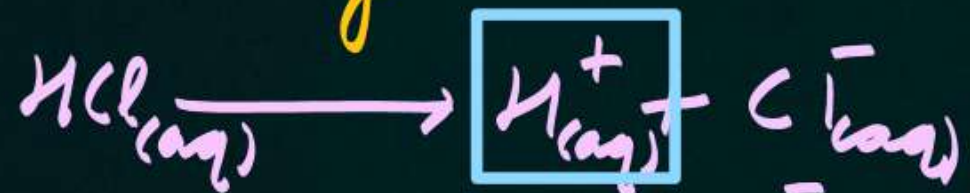
b) BROWNSTED-LOWRY CONCEPT.

c) LEWIS CONCEPT.

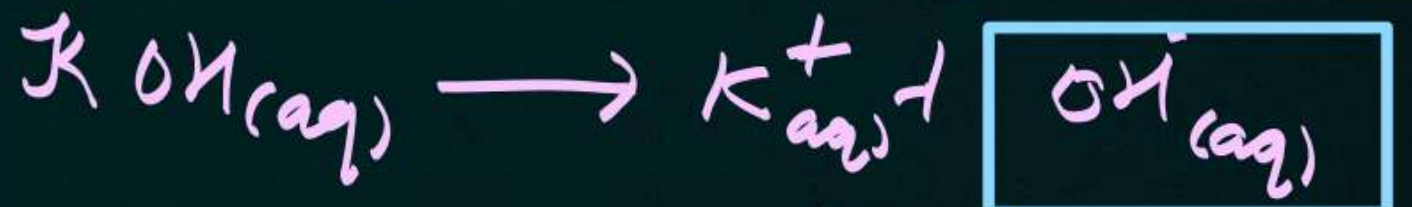
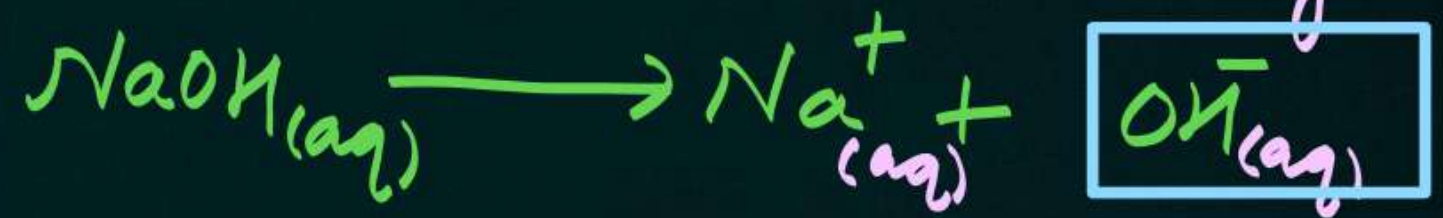
Concept of Acids and Bases

1. Arrhenius Concept of Acids and Bases

ACID $\rightarrow$ According to this concept substances that produce H^+ in water are called acid. e.g.



BASE $\rightarrow$ According to this concept substances that produce OH^- ions in water are called Base. e.g.

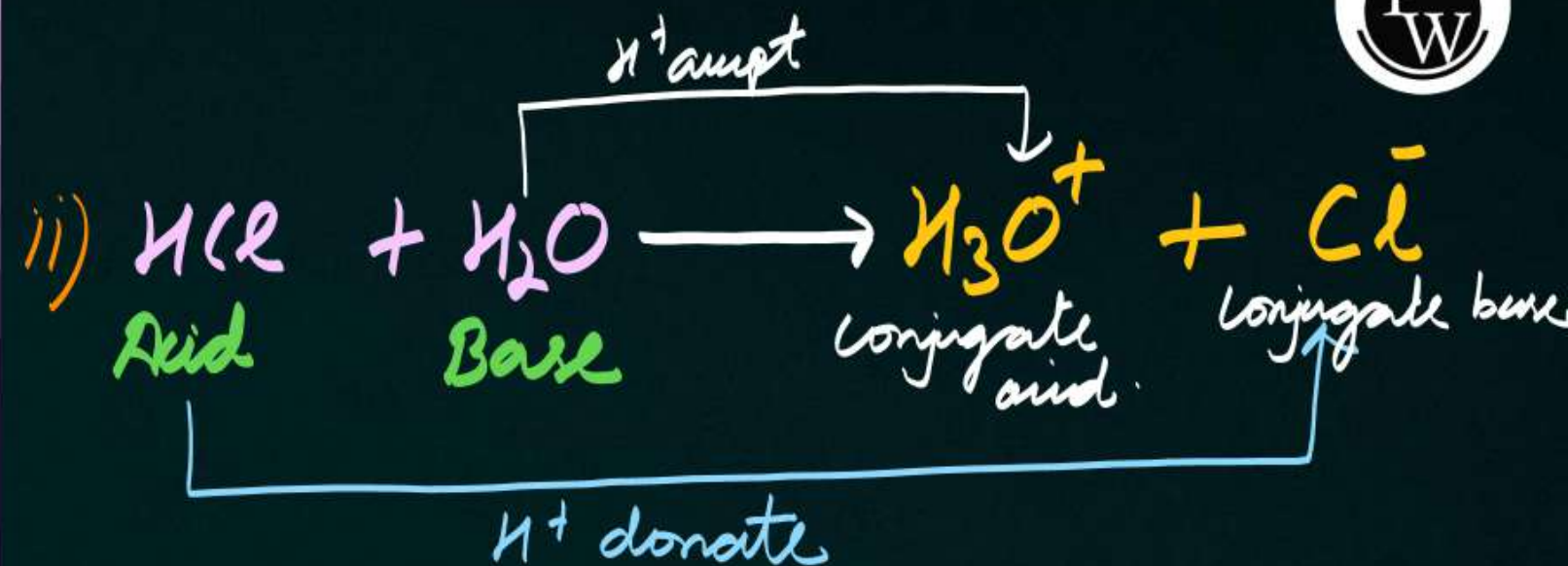
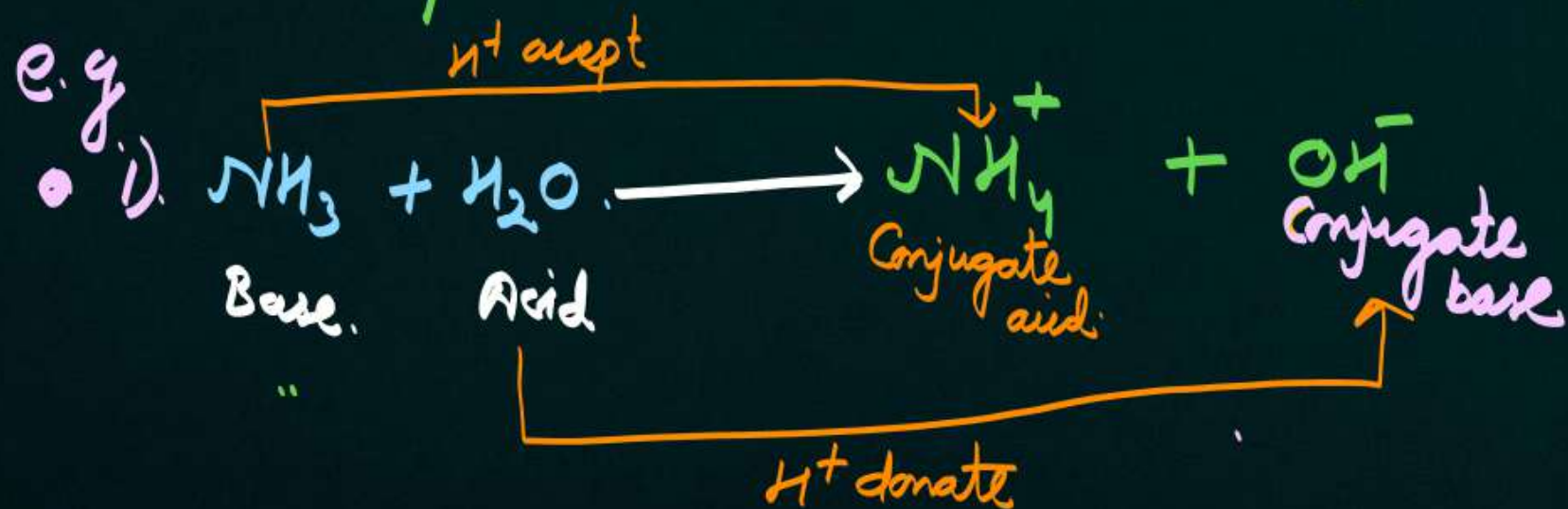


Concept of Acids and Bases

2. The Brönsted-Lowry Acids and Bases

Acid $\rightarrow$ According to this concept substances that donate H^+ are called acids.

Base $\rightarrow$ According to this concept substances that accept H^+ ions are called base.



- That means same compound can be behave as acid as well as base.
- Strong acid after donating H^+ become weak base & vice versa.
- Strong base after accepting H^+ become weak acid & vice versa.
- Conjugate Acid & base pair has difference of only one H^+ .



Species	Conjugate acid	Conjugate acid ^{base}
$\text{H}_2\text{O} \xrightarrow{+\text{H}^+} \text{H}_3\text{O}^+$	H_3O^+	OH^-
$\text{HCO}_3^- \xrightarrow{+\text{H}^+}$	H_2CO_3	CO_3^{2-}
$\text{HSO}_4^- \xrightarrow{+\text{H}^+}$	H_2SO_4	SO_4^{2-}
$\text{NH}_3 \xrightarrow{+\text{H}^+}$	NH_4^+	NH_2^-

QUESTION

What will be the conjugate bases for the following Bronsted acids?

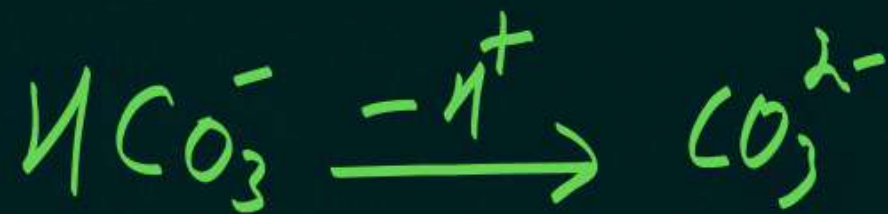
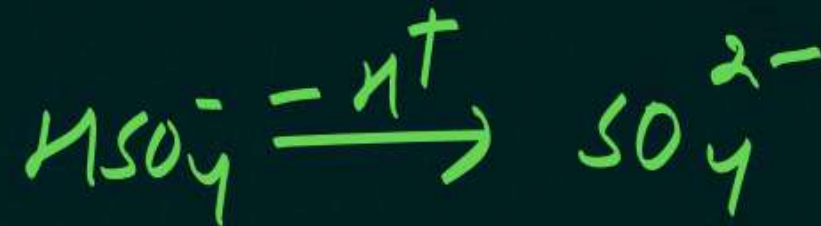
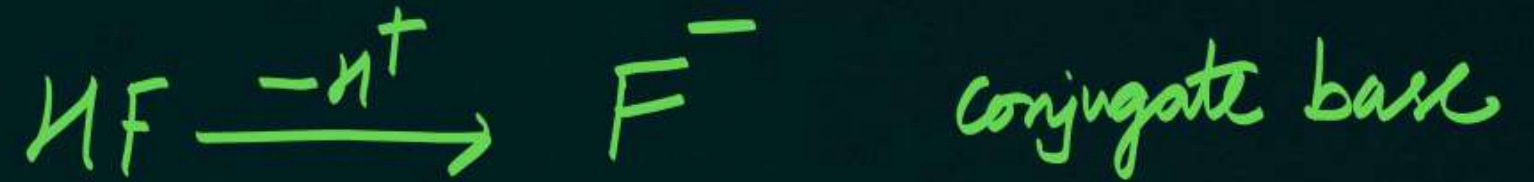
HF, HSO₄⁻ and HCO₃⁻

1 F⁻, SO₄²⁻ and CO₃²⁻

2 F⁻, SO₄²⁻ and H₂CO₃

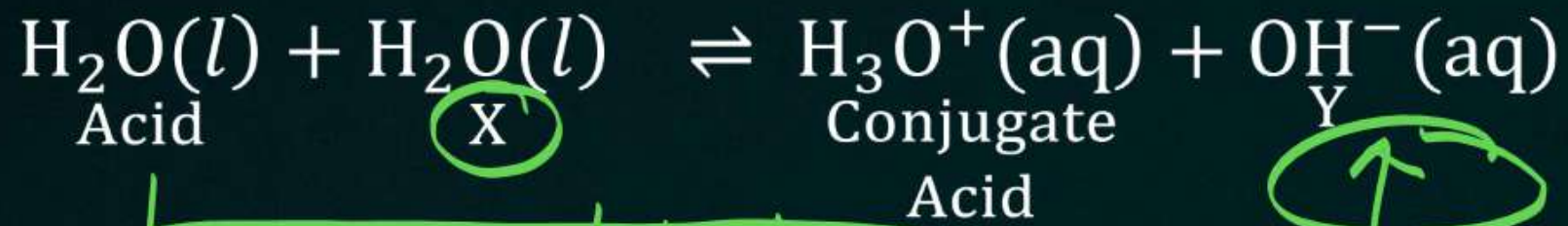
3 F⁻, HSO₄⁻ and H₂CO₃

4 F⁻, HSO₄⁻ and CO₃²⁻



QUESTION

In the following reaction,



X and Y respectively are

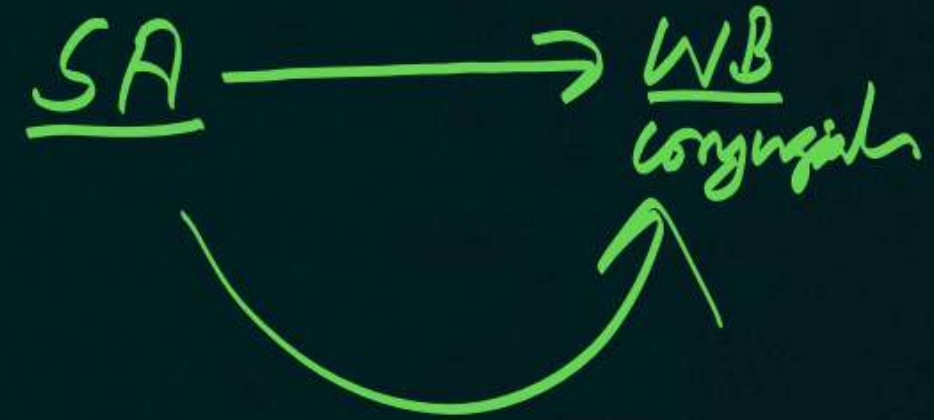
Lalnu

- 1 Base and conjugate acid
- 2 Acid and conjugate base 
- 3 Base and conjugate base
- 4 Acid and conjugate acid 

QUESTION

Strong acid dissociates completely in water, the resulting base formed would be very weak. The reason is that

- 1 Strong acids have strong conjugate bases
- 2 Strong acids have strong conjugate acids
- 3 Strong acids have very weak conjugate bases
- 4 Strong acids have very weak conjugate acids



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② Wahid

③ Sheikh Ayman.

REVISION ZARUR KARNA.

Thank
You